Annexure-2

Guidelines for construction of Piezometers and monitoring of Ground Water Levels and Quality

A piezometer is a borewell/ tube well used only for measuring the water level by lowering a tape/ sounder or automatic/ digital water level measuring equipment. It also takes water samples for water quality testing whenever needed. General guidelines for installation of piezometers are as follows:

1. The Piezometer is to be installed/ constructed at a minimum distance of 50 m from the pumping well through which groundwater is being withdrawn. The diameter of the Piezometer should be about four inches to six inches.
2. The depth of the Piezometer should be the same as that of the pumping well from which groundwater is being abstracted. If more than one pumping well is constructed to tap aquifers at different depths, more than one Piezometer shall be required to be constructed to tap different aquifers, as in the pumping wells.
3. The measurement of the water level in the Piezometer should be taken only after the pumping from the surrounding tube wells has been stopped for about four to six hours.
4. The groundwater quality has to be monitored once a year during the pre-monsoon (April/May) period by industries and mines drawing groundwater. Samples of groundwater should be analyzed from an accredited laboratory.
5. A permanent display board should be installed at the Piezometer/ Tube well site to provide the location, piezometer/ tube well number, depth and zone tapped of the piezometer/ tube well for standard referencing and identification.
6. Any other site-specific requirement regarding safety and access for measurement may be taken care off.